How-To Library

Informatica

Address Verification 5.x Best Practices for Canada

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Abstract

This document tells you how to connect the elements in a Canada address to the appropriate address elements in Informatica Address Verification. This document also describes best practices that you can implement when you verify addresses in Canada.

Supported Versions

- Informatica Address Verification 3.0 and later
- Informatica Address Verification (On-Premises) 5.x

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Introduction

Informatica Address Verification can validate and enhance addresses in Canada. This how-to article provides you with configuration information and best practices that you can follow to get the most from the addresses. The best practices include advice on how to select the appropriate XML elements for Canada address data.

To verify the quality of an input address, Address Verification compares the address information to the data in one or more reference data files. You can configure Address Verification to identify the different types of address information in the input data.

When you select the input address elements, you must ensure that the elements are correct for the different types of input information. The country that the address identifies determines the elements that you need.

An address in Canada can contain different types of information, including street information and locality information. You must map each item of address information to an appropriate element in Address Verification.

Process Modes

Informatica Address Verification supports the following process modes for Canada addresses:

- BATCH
- CERTIFIED
- COUNTRY RECOGNITION
- FAST COMPLETION

- INTERACTIVE
- PARSE

Character Set

Canada reference address databases store address data in the Latin-1 script. By default, Informatica Address Verification returns the verified Canada addresses in the Latin-1 script.

Address Format and Address Structure

You improve the performance of Address Verification when you define the input address in the format that the postal service expects.

The Canada postal service recognizes the following structure for a postal address in English language:

```
Line 1: RECIPIENT OR CONTACT NAME
Line 2: [SUB-BUILDING NUMBER -] HOUSE OR BUILDING NUMBER STREET NAME STREET DESC STREET
DIRECTION
Line 3: CITY PROVINCE POSTAL CODE
Line 4: COUNTRY
```

The following sample address illustrates the structure:

```
811 PORTAGE AVE
WINNIPEG MB R3G 0N3
CAN
```

The Canada postal service recognizes the following structure for a postal address in French language:

```
Line 1: RECIPIENT OR CONTACT NAME
Line 2: [SUB-BUILDING NUMBER -] HOUSE OR BUILDING NUMBER STREET DESC STREET NAME STREET
DIRECTION
Line 3: CITY PROVINCE POSTAL CODE
Line 4: COUNTRY
```

The following sample address illustrates the structure:

```
811 AV PORTAGE
WINNIPEG MB R3G 0N3
CAN
```

Key Address Elements

Address Verification is unlikely to verify an address successfully if one of the following items is absent from the address:

- Street
- Locality
- Postal Code

Mapping Input Address Data to Address Elements

To verify an address, map the input address information to appropriate elements in Address Verification.

The following table shows how you can map the input address to the address elements:

Information Type	Address Information	Address Elements
Street Name, Street Type	PORTAGE AVE	Street
House Number	811	HouseNumber
Postal Code	R3G 0N3	PostalCode
City	WINNIPEG	Locality
State	MB	ProvinceAbbreviation

Input and Result XML in Address Verification (On-Premises)

The following example shows how you can map the address information to the XML address elements in Address Verification (On-Premises).

You enter the following input address:

```
<InputData>
    <AddressElements>
        <Country Item="1" Type="NAME">CANADA</Country>
        <Locality Item="1" Type="COMPLETE">WINNIPEG</Locality>
        <PostalCode Item="1" Type="UNFORMATTED">R3G 0N3</PostalCode>
        <Province Item="1" Type="COUNTRY_STANDARD">MB</Province>
        <Street Item="1" Type="COMPLETE">811 PORTAGE AVE</Street>
    </AddressElements>
</InputData>
```

Informatica Address Verification (On-Premises) returns the following result:

```
<?xml version="1.0" encoding="UTF-16"?>

<!-- EngineVersion="5.11.0.41659" -->

<Result     ProcessStatus="V4"
     ModeUsed="INTERACTIVE"
     Count="1"
     CountOverflow="NO"
     CountryISO3="CAN"
     PreferredScript="LATIN_1"
     PreferredLanguage="ENGLISH">

<ResultData ResultNumber="1"
     MailabilityScore="5"
     ResultPercentage="100.00"
     ElementResultStatus="F0F0F0F0F000000000E0"
     ElementInputStatus="60606060600000000060"
     AddressResolutionCode="0000000000000000000000"
     ExtElementStatus="00000000000000000000"
     ElementRelevance="10101010100000000010"
     AddressType="S"
     LanguageISO3="ENG">
    <AddressElements>
        <Country Type="NAME_EN" Item="1">Canada</Country>
        <Locality Item="1">Winnipeg</Locality>
        <PostalCode Item="1">R3G 0N3</PostalCode>
        <Province Item="1">MB</Province>
        <Street Item="1">Portage Ave</Street>
```

```
            <Number Item="1">811</Number>
        </AddressElements>
        <AddressLines>
            <DeliveryAddressLine Line="1">811 Portage Ave</DeliveryAddressLine>
            <CountrySpecificLocalityLine Line="1">Winnipeg MB R3G 0N3</CountrySpecificLocalityLine>
            <FormattedAddressLine Line="1">811 Portage Ave</FormattedAddressLine>
            <FormattedAddressLine Line="2">Winnipeg MB R3G 0N3</FormattedAddressLine>
            <FormattedAddressLine Line="3">Canada</FormattedAddressLine>
        </AddressLines>
        <AddressComplete>811 Portage Ave
     Winnipeg MB R3G 0N3
     Canada</AddressComplete>
        </ResultData>
    </Result>
```

Input and Result XML in Address Verification

The following example shows how you can verify a Canada address in a SOAP call to Address Verification.

You submit the following request:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns="http://
validator5.AddressDoctor.com/Webservice5/v4">
 <soap:Header/>
 <soap:Body>
   <Process>
    <login>****</login>
    <password>****</password>
    <parameters>
      <ProcessMode>BATCH</ProcessMode>
    </parameters>
    <ValidationParameters>
        <AdditionalInformationSet>
           <Name>EngineVersion</Name>
           <Value>ON</Value>
        </AdditionalInformationSet>
    </ValidationParameters>
    <addresses>
     <Address>
       <Country>
         <string>CANADA</string>
       </Country>
       <FormattedAddress>
           <string>811 PORTAGE AVE</string>
           <string>WINNIPEG MB R3G 0N3</string>
       </FormattedAddress>
     </Address>
    </addresses>
    <enrichments></enrichments>
   </Process>
 </soap:Body>
</soap:Envelope>
```

Informatica Address Verification returns the following response:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
 <soap:Body>
    <ProcessResponse xmlns="http://validator5.AddressDoctor.com/Webservice5/v4">
       <ProcessResult>
          <StatusCode>100</StatusCode>
          <StatusMessage>OK</StatusMessage>
          <Results>
             <Result>
                <ProcessStatus>V4</ProcessStatus>
                <CountryISO3>CAN</CountryISO3>
                <ResultDataSet>
                   <ResultData>
                      <ResultNumber>1</ResultNumber>
                      <MailabilityScore>5</MailabilityScore>
                      <ResultPercentage>100.00</ResultPercentage>
                      <ElementInputStatus>6060606060000000060</ElementInputStatus>
```

```
                        <ElementResultStatus>F0F0F0F0F000000000E0</ElementResultStatus>
                        <ElementRelevance>1010101010000000010</ElementRelevance>
                        <ExtElementStatus>00000050000000000000</ExtElementStatus>
                        <AddressResolutionCode>00000000000000000000</AddressResolutionCode>
                        <AddressType>S</AddressType>
                        <LanguageISO3>ENG</LanguageISO3>
                        <Address>
                           <Street>
                              <string>PORTAGE AVE</string>
                           </Street>
                           <HouseNumber>
                              <string>811</string>
                           </HouseNumber>
                           <Locality>
                              <string>WINNIPEG</string>
                           </Locality>
                           <PostalCode>
                              <string>R3G 0N3</string>
                           </PostalCode>
                           <Province>
                              <string>MB</string>
                           </Province>
                           <Country>
                              <string>CANADA</string>
                           </Country>
                           <DeliveryAddressLines>
                              <string>811 PORTAGE AVE</string>
                           </DeliveryAddressLines>
                           <CountrySpecificLocalityLine>
                              <string>WINNIPEG MB R3G 0N3</string>
                           </CountrySpecificLocalityLine>
                           <FormattedAddress>
                              <string>811 PORTAGE AVE</string>
                              <string>WINNIPEG MB R3G 0N3</string>
                              <string>CANADA</string>
                           </FormattedAddress>
                           <AddressComplete>811 PORTAGE AVE;WINNIPEG MB R3G 0N3;CANADA</
AddressComplete>
                        </Address>
                     </ResultData>
                  </ResultDataSet>
               </Result>
            </Results>
         </ProcessResult>
      </ProcessResponse>
   </soap:Body>
</soap:Envelope>
```


Address Enrichments

You can configure Informatica Address Verification to add enrichments to verified Canada addresses.

The following table lists the type of enrichments that you can add to a Canada address:

Enrichment Type	Description
CAMEO	You can retrieve CAMEO profiles as enrichments for valid Canada addresses. CAMEO profiles are consumer classification systems that provide socio-economic and demographic indicators about the residents at the address.
Geocoding	You can retrieve the standard, interpolated geocoordinates for valid Canada addresses.

Enrichment Type	Description
Arrival Point Geocoding	You can retrieve the arrival point geocoordinates for valid Canada addresses.
Parcel Centroid Geocoding	You can retrieve the parcel centroid geocoordinates for valid Canada addresses.

Frequently Asked Questions

Can Address Verification return address information in multiple languages for Canada?

You can configure Address Verification to return valid Canada address information in English or in French.

English is the default language for addresses from all provinces except Quebec. French is the default language for Quebec addresses.

To set the preferred language, configure the `PreferredLanguage` attribute of the Result element in `Parameters.xml`.

The following table describes the `PreferredLanguage` values that you can set for Canada address data:

Option	Description
DATABASE	Default value. Returns addresses in English for all provinces except Quebec. Returns Quebec addresses in French.
ENGLISH	Returns all address elements in English.
ALTERNATIVE_1	Returns all addresses in English.
ALTERNATIVE_2	Returns Quebec addresses in French. In provinces other than Quebec, returns the street descriptors, directional information, and province names in French. Returns other address elements in English.
PRESERVE_INPUT	Returns the address in the input language.

How does Address Verification handle sub-building information in different process modes?

The batch and interactive reference address databases contain Postal Code Address Data (PCAD). When you verify an address in batch or interactive mode, Address Verification reads the PCAD data.

The PCAD file might not include sub-building data for all primary addresses that specify the data. In such cases, Address Verification can return range-based sub-building information for the addresses. In batch mode, Address Verification verifies an address if the input sub-building data matches the sub-building data in the reference data. In interactive mode, Address Verification can return address suggestions with sub-building data if the input address data supports one or more matches to sub-building data in the reference data.

The certified reference address databases contain Point of Call Address Data (PoCAD). When you verify an address in certified mode, Address Verification reads the PoCAD data.

As per the Software Evaluation and Recognition Program (SERP), certified mode can return exact sub-building information for most primary addresses. In certified mode, Address Verification rejects an address with valid street information if the required sub-building information is not provided in the input address.

How does Address Verification handle P.O. Box, Station, and Rural Route information in Canada?

Address Verification returns the P.O. Box, Station, and Rural Route information in the DeliveryService field.

When you submit an address, provide the P.O. Box, Station, and Rural Route information in the DeliveryService field.

Can Address Verification verify an address with minimum address information when an input address contains a Large Volume Receiver (LVR) postal code?

Yes. Address Verification can verify an address with minimum address information if the input address contains an LVR postal code.

For example, consider an input address with an LVR postal code and country data:

```
<InputData>
  <AddressElements>
      <Country Item="1" Type="NAME">can</Country>
      <PostalCode Item="1" Type="UNFORMATTED">M5J 2J5</PostalCode>
  </AddressElements>
</InputData>
```

Address Verification validates and returns the P.O. Box address for the LVR postal code:

```
<!-- EngineVersion="5.12.0.42716" -->

<Result ProcessStatus="C4"
                ModeUsed="BATCH"
                Count="1"
                CountOverflow="NO"
                CountryISO3="CAN"
                PreferredScript="DATABASE"
                PreferredLanguage="DATABASE">

<ResultData ResultNumber="1"
                MailabilityScore="4"
                ResultPercentage="100.00"
                ElementResultStatus="F08080000088000000E0"
                ElementInputStatus="6000000000000000060"
                AddressResolutionCode="00000000000000000000"
                ExtElementStatus="00500000000000000000"
                ElementRelevance="10101000001100000010"
                AddressType="L"
                LanguageISO3="ENG">
                <AddressElements>
                                <Country Type="NAME_EN" Item="1">CANADA</Country>
                                <Locality Item="1">TORONTO</Locality>
                                <PostalCode Item="1">M5J 2J5</PostalCode>
                                <Province Item="1">ON</Province>
                                <DeliveryService Item="1">PO BOX 1</DeliveryService>
                                <DeliveryService Item="2">STN ROYAL BANK</
DeliveryService>
                </AddressElements>
```

Reference Address Databases

Informatica updates the Canada reference address databases monthly. You can download the latest database packages from the Informatica Database Download portal. For news about reference address databases and software updates for Address Verification, subscribe to the UpData newsletter from Informatica.

Informatica Address Verification provides the following reference databases for Canada addresses:

- `CAN5BI.MD`. Batch and interactive modes.
- `CAN5C1.MD`. Certified mode.
- `CAN5FC.MD`. Fast completion mode.

- `CAN5GC.MD`. Standard geocoding.
- `CAN5GCAP.MD`. Arrival point geocoding.
- `CAN5GCPC.MD`. Parcel centroid geocoding.
- `CAN5CA.MD`. CAMEO data.

Note: You download databases for Informatica Address Verification (On-Premises). You do not need to download databases for Informatica Address Verification, as Informatica Address Verification runs on Informatica Cloud.

Additional Documentation Resources

To view the latest documentation for Informatica Address Verification, click the following links:

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor-cloud

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